

Target-site resistance to acetolactate synthase (ALS)-inhibiting herbicides in *Amaranthus palmeri* from Argentina

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Abstract

BACKGROUND: Herbicide-resistant weeds are a serious problem worldwide. Recently, two populations of *Amaranthus palmeri* with suspected cross-resistance to acetolactate synthase (ALS)-inhibiting herbicides (R1 and R2) were found by farmers in two locations in Argentina (Vicuña Mackenna and Totoras, respectively). We conducted studies to confirm and elucidate the mechanism of resistance.

RESULTS: We performed *in vivo* dose–response assays, and confirmed that both populations had strong resistance to chlorimuron-ethyl, diclosulam and imazethapyr when compared with a susceptible population (S). *In vitro* ALS activity inhibition tests only indicated considerable resistance to imazethapyr and chlorimuron-ethyl, indicating that other non-target mechanisms could be involved in diclosulam resistance. Subsequently, molecular analysis of *als* nucleotide sequences revealed three single base-pair mutations producing substitutions in amino acids previously associated with resistance to ALS inhibitors, A122, W574, and S653.

CONCLUSION: This is the first report of *als* resistance alleles in *A. palmeri* in Argentina. The data support the involvement of a target-site mechanism of resistance to ALS-inhibiting herbicides.

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Keywords: weeds; *Amaranthus palmeri*; herbicide resistance; acetolactate synthase; acetolactate synthase inhibitors; cross-resistance

1 INTRODUCTION

For centuries, growing plants as food and fiber sources has been a major human endeavor. At present, with a world population over 7.5 billion people, any threat to production systems directly impacts global sustainability. One of these threats, if not the main one, remains the infestation of crops by wild plants with powerful adaptive and propagative abilities.¹ In the last four decades, herbicides have, by far, been the main strategy used for weed control, largely replacing hand weeding and mechanical control tools.² Although herbicides contributed substantially to the high productivity of global agriculture, they did not result in the complete control of weeds. Instead, monoculture production systems and repeated use of herbicides with similar mechanisms of action have led to herbicide resistance in weeds.^{3–6} Worldwide, there are currently 252 weed species with biotypes resistant to one or more herbicides, and resistance has evolved for 23 of the 26 known herbicide sites of action.⁷

Palmer amaranth (*Amaranthus palmeri* S. Wats.) is one of the approximately 60 *Amaranthus* species native to the Americas,^{8,9} and is the most troublesome for the cultivation of soybean (*Glycine max* L. Merr.), corn (*Zea mays* L.) and cotton (*Gossypium hirsutum* L.), all on the American continent.^{10–16} *Amaranthus* species are characterized by having an extended period of germination, rapid growth, and prolific seed production.^{17,18} Palmer amaranth is an annual dioecious broadleaf weed capable of growing 3 to 4 m

tall¹⁹ and produces unbranched terminal seedheads up to 0.5 m in length,²⁰ with seed production of up to 600 000 seeds per female plant.²¹ Moreover, compared with redroot pigweed (*Amaranthus retroflexus* L.), common waterhemp (*Amaranthus rudis* S.), and tumble pigweed (*Amaranthus albus* L.), Palmer amaranth has shown the highest values for plant volume, dry weight, and leaf area.²² Palmer amaranth has a long dormant seed period, a rapid growth, and the potential ability to acquire herbicide resistance by successfully outcrossing with other species from the same genus.¹² Consequently, this weed detrimentally affects crop growth and yield by effectively enhancing its competitiveness for resources like light, water, space, and nutrients.²³

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The first report of herbicide resistance in Palmer amaranth came in 1989 in South Carolina, USA.²⁴ Since then, the high selection pressure of modern farming systems has led to the increasing emergence of cases of resistance in this species, covering as many as six different mechanisms of action [5-enolpyruvylshikimate-3-phosphate synthase (EPSPS); aceto-lactate synthase (ALS); photosystem II; 4-hydroxyphenylpyruvate dioxygenase (HPPD); microtubule inhibitors; and protoporphyrinogen oxidase (PPO) inhibitors], and biotypes developing multiple resistance to two or three of the mechanisms mentioned.⁷ In Argentina, mucronate amaranth (*Amaranthus quitensis* L.) is largely considered the most problematic weed of recent times, with no official reports of the presence of *A. palmeri* until 2013.^{13,25} Given the high tendency of *A. palmeri* to acquire resistance to several mechanisms of action, it is highly relevant to characterize the reported field resistances and carry out detailed studies of the molecular basis of these resistances. This information will be useful in designing control strategies that allow farmers to prevent, understand and overcome this weed resistance problem without creating a similar one in the near future. Hence, the objectives of this work were to confirm the resistance of two suspected cross-resistant *A. palmeri* populations found in two Argentinian farms, and to characterize that resistance at the molecular level.

2 MATERIALS AND METHODS

2.1 Plant material and seed collection

Seeds of Palmer amaranth were collected from three farms, sampling at least 100 plants combined into a single composite sample per field. At two of these farms, it was known that *A. palmeri* had survived herbicide applications of Derby® (a.i. imazethapyr 10% w/v; Gleba, La Plata, Argentina), and at the third farm the population was susceptible to this herbicide's mode of action.

The populations with suspected cross-resistance to ALS herbicides were named R1 (from Vicuña Mackenna; 33° 55' 20.68" S, 64° 35' 33.37" W) and R2 (from Totoras; 32° 34' 59.88" S, 61° 10' 59.88" W); and the susceptible population from Tucumán (27° 17' 45.36" S, 65° 0' 3.37" W) was referred to as S.

2.2 Chemical compounds

For both *in vitro* and *in vivo* dose–response assays, three herbicides were tested: Derby® (a.i. imazethapyr 10% w/v; Gleba), VRILEC® (a.i. chlorimuron-ethyl 25% w/w; Agrofina, Buenos Aires, Argentina), and Spider® (a.i. diclosulam 84% w/w; Dow Agro-Sciences, Indianapolis, IN, USA). The active ingredients correspond to the ALS-inhibitor chemical families: imidazolinones (IMIs), sulfonylureas (SUs) and triazolopyrimidines (TPs), respectively.

2.3 In vivo dose–response assays

To assess each herbicide effect, approximately 120 seedlings of the resistant and susceptible populations were grown in 9-cm plastic pots containing a mixture of soil, sand and perlite (70–20–10%, respectively), previously sterilized with methyl bromide. The experiment was conducted in a randomized complete block design with five replications per dose, using three plants per replication. When plants reached the stage of four to six true leaves, different doses of herbicides were sprayed onto them, depending on the population, in order to obtain a good curve fit in each case. The following doses were used: 0, 0.0625X, 0.125X, 0.250X, 0.5X, 1X, 2X, 4X, 8X, 16X and 32X, with X = 100 g a.i. ha⁻¹ for imazethapyr, 50 g a.i. ha⁻¹ for chlorimuron-ethyl, and 30 g

Table 1. Primers used in polymerase chain reactions

Name	Sequence (5' – 3')
X1	CTTCAAGCTTCAACAATGG
X2	CCTACAAAAGCTTCTCTCT
X3	CCAGAAAAGGTTGCGATGTT
X4	ATGCTTGGAATGGATCGAG
X5	GGCAACCTAGAAAGATACCC
X6	CCTGAATGGCGTATTGCGGAGG
X7	GGAGGGAGGAGTTGAATGAGCAG

a.i. ha⁻¹ for diclosulam. These doses correspond to the recommended label rates. Herbicides were applied using a constant pressure backpack with flat fan nozzles at 206.84 kPa calibrated to deliver 75 L ha⁻¹. Three weeks after treatment, the number of survivors was recorded. Dry weight was recorded by harvesting the above-ground plant tissue, placing it in a paper bag and drying it at 60 °C for 48 h. Data were expressed as percentage of plant survival and biomass reduction relative to the untreated control.

2.4 In vitro enzyme activity assays

Fresh leaves of six plants, 21 days after germination, of both the resistant and susceptible populations were used to extract proteins according to Poston *et al.*²⁶ and enzyme activity was determined by measuring the amount of acetoin formed as a function of the doses of different herbicides tested, according to the method of Westerfeld, modified by Yu *et al.*²⁷ This method allows the formation of a pink complex in a medium that also contains creatine and α -naphthol. The intensity of the color was quantified by measuring absorbance at 490 nm. Herbicide concentrations used for enzyme activity testing were 0, 0.25, 1, 5, 250 and 1000 μ M for imazethapyr, and 0, 0.025, 0.1, 0.25, 1, 5, 25 and 250 μ M for chlorimuron-ethyl and diclosulam. Enzyme inhibition was computed as I_{50} , which represents the herbicide concentration required to reduce the enzyme activity by 50%, compared with the untreated control. There were three replications per treatment per population, and experiments were conducted three times.

2.5 Cloning and sequencing of the *als* gene

Genomic DNA was extracted from fresh leaf tissue of ten plants of each population (S, R1 and R2) using the Wizard Genomic Extraction Kit® (Promega Corp., Madison, WI, USA), and genomic bulks were used as a template for *als* gene amplification with primers designed from *als* sequences of related species published in available database.^{28,29} The complete *als* gene was amplified with primers X1 and X2 (Table 1) using Q5® High-Fidelity DNA polymerase (New England Biolabs, Ipswich, MA, USA). Reactions consisted of ~100 ng of DNA, 500 nM primers, 1.5 mM MgCl₂, 200 μ M dNTPs, 1 $\times$ buffer, 1 polymerase unit, and H₂O to 50 μ L. Cycle conditions were as follows: 98 °C for 1 min, 35 cycles of 98 °C for 10 s, 57 °C for 30 s, and 72 °C for 1 min, 72 °C for 7 min, and a 12 °C hold. Amplification products were ligated to the pGEM®-T easy vector (Life Technologies, Grand Island, NY, USA) as described in the technical manual. Chemically competent DH5 α *Escherichia coli* cells were prepared and transformed according to Sambrook and Russell³⁰ and transformants were selected for isolation on Luria-Bertani plus 100 μ g mL⁻¹ ampicillin plates. Insert integration was checked by carrying out a specific colony polymerase chain reaction (PCR), using primers

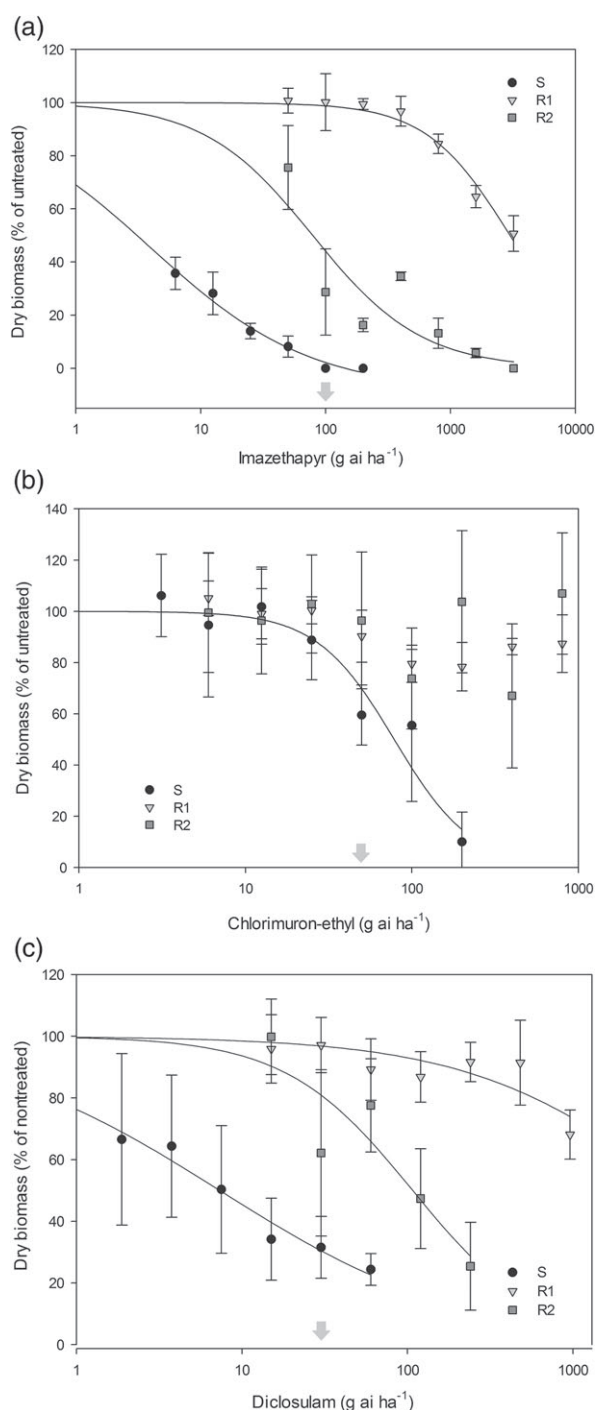


Figure 1. *In vivo* dose–response (dry biomass) curves with ALS-inhibiting herbicides in *A. palmeri*. Plants of the R1, R2 and S populations were subjected to the application of different doses of imazethapyr (a), chlorimuron-ethyl (b) and diclosulam (c). Results are expressed as a percentage of remaining biomass in comparison to untreated control groups. Arrows indicate doses applied to the field.

X3 and X4 (Table 1). Overnight cultures of positive colonies were used to make a plasmid miniprep, according to the Wizard® Plus SV Minipreps DNA Purification System protocol (Promega Corp.). Eight positive-recombinant DNA molecules from each population were sequenced (Maregen Inc., Seoul, Korea), employing the T7 and SP6 primers for the pGEM-T vector backbone and the X5, X6, and X7 primers designed to cover the

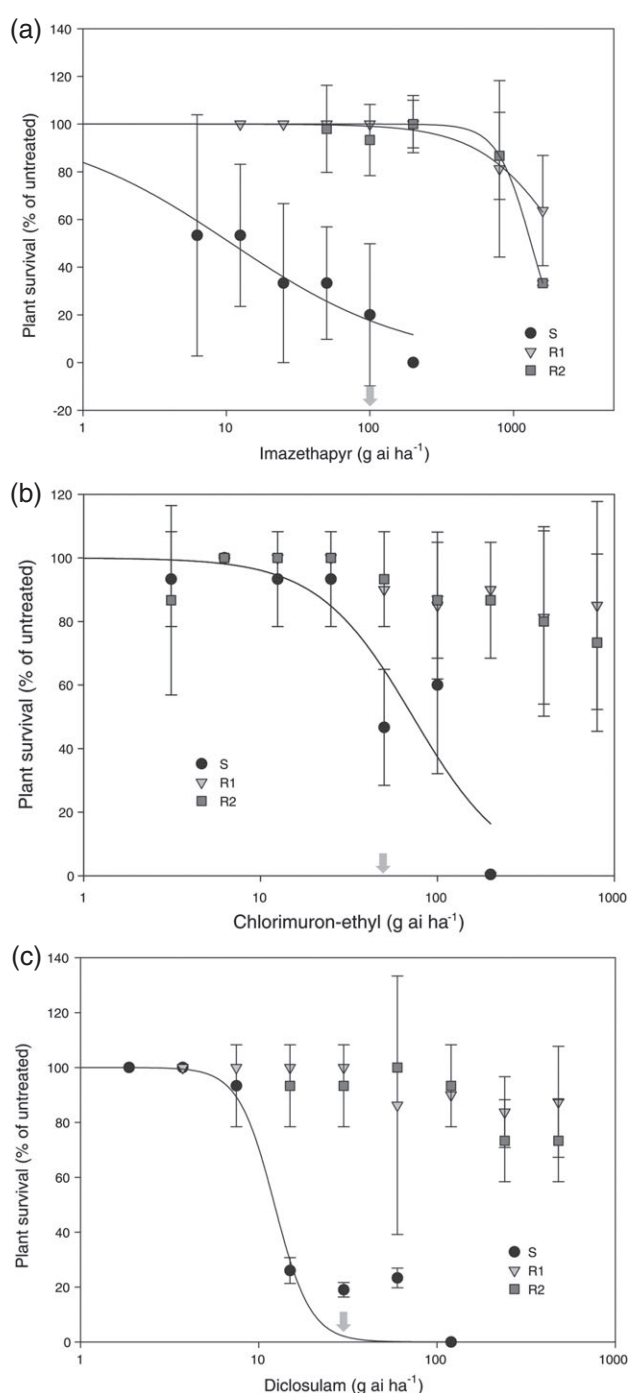


Figure 2. *In vivo* dose–response (plant survival) curves with ALS-inhibiting herbicides in *A. palmeri*. Plants of the R1, R2 and S populations were subjected to the application of different doses of imazethapyr (a), chlorimuron-ethyl (b) and diclosulam (c). Results are expressed as a percentage of plant survival in comparison to untreated control groups. Arrows indicate doses applied to the field.

entire *als* gene sequence (Table 1). This cloning strategy allows the unambiguous determination of the complete sequence of individual *als* alleles.

2.6 *als* sequence analysis

To manually generate complete R1, R2 and S *als* sequences, sequencing reads were aligned with reference *als* sequences

Table 2. Dose–response parameters

Herbicide	Population	Dry weight			Plant survival		
		GR ₅₀	Rf		LD ₅₀	Rf	
Imazethapyr	R1	2996	± 206 ^a	788	2288	± 135 ^a	211
	R2	78.8	± 25.9 ^b	21	1327	± 50 ^b	123
	S	3.8	± 0.6 ^c		10.82	± 2.82 ^c	
Chlorimuron-ethyl	R1	>800	>9.3		>800		>11
	R2	>800	>9.3		>800		>11
	S	86.2	± 12.5		73 ± 17		
Diclosulam	R1	4397	± 3214 ^a	600	>960		>78
	R2	128	± 39 ^b	17	>960		>78
	S	7.32	± 0.82 ^c		12.2 ± 1.5		

GR₅₀/LD₅₀ units are g a.i. ha⁻¹. Resistance factors (Rfs) were calculated using concentrations of inhibitors required to reduce dry weight or plant survival by 50%. For each herbicide, parameters with different letters indicate statistically significant differences.

Table 3. Parameters of ALS activity fit curves

Herbicide	Population	Parameters	
		I ₅₀ (μM)	Rf
Imazethapyr	R1	17.50 ± 9.09 ^a	33.68
	R2	21.98 ± 6.23 ^a	42.30
	S	0.5196 ± 0.3418 ^b	
Chlorimuron-ethyl	R1	0.0098 ± 0.0020 ^a	6.53
	R2	0.0145 ± 0.0054 ^a	9.67
	S	0.0015 ± 0.0004 ^b	
Diclosulam	R1	0.0136 ± 0.0032 ^a	2.27
	R2	0.0171 ± 0.0029 ^a	2.85
	S	0.0060 ± 0.0006 ^b	

Resistance factors (Rfs) were calculated using concentrations of inhibitors required to reduce ALS activity by 50% (I₅₀). Resistance factors can only be compared within a chemical family. For each herbicide, I₅₀ values with different letters indicate statistically significant differences.

from the *Amaranthus* genus obtained from the National Center for Biotechnology Information (NCBI). The alignment was performed using the MUSCLE algorithm from Unipro Ugene v1.11.2 software.³¹ All sequences were translated and aligned, to search for amino acid substitutions.

2.7 Statistical analysis

Dose–response curves were obtained with a nonlinear log-logistic regression model using SigmaPlot (version 11.0; Systat Software, Inc., San Jose, CA, USA) according to the following equation:

$$y = \min + \frac{(\max - \min)}{1 + \left(\frac{x}{b}\right)^a}$$

where y is the percentage of dry weight or plant survival, x is the herbicide concentration (rate), $\min$ and $\max$ are the lower and the upper limits of the curve, respectively, a is the slope at the inflexion point, and b is the herbicide dose rate required to reduce dry weight by 50% [50% growth reduction (GR50)] or to reduce plant survival by 50% [50% lethal dose (LD50)].

A different nonlinear regression analysis was applied to fit ALS activity data to a sigmoidal three parametric logistic curve using SigmaPlot (version 11.0; Systat Software, Inc.) according to the following equation:

Table 4. Amino acid differences found in the ALS open reading frame of resistant *Amaranthus palmeri* populations compared with the susceptible population

Population	Allelic version (and frequency)	ALS substitution	Accession number
S	A (8/8)	–	KY781916
R1	A (1/8)	–	
	B (5/8)	A282D	KY781917
	C (1/8)	W574 L	KY781918
R2	D (1/8)	W574 L – M601I	KY781919
	B (2/8)	A282D	
	E (2/8)	A122S – A282D	KY781920
	F (1/8)	A282D – S653 N	KY781921
	G (1/8)	P84H – A282D – S653 N	KY781922
	H (2/8)	S653 N	KY781923

Previously reported substitutions are in bold. Amino acids are numbered according to the *Arabidopsis thaliana* ALS sequence, with the exception of A282D, which is numbered according to the *Amaranthus retroflexus* ALS sequence, as it falls in an indel

$$y = \frac{a}{1 + \left(\frac{x}{150}\right)^b}$$

where y is the ALS activity, x is the herbicide concentration (rate), a is the upper response limit, b is the slope of the curve to relate the effect of herbicide dose and concentration, and I_{50} is the herbicide concentration required to cause a 50% reduction in ALS enzyme activity.

Data from the experiments were tested using one-way analysis of variance (ANOVA). Minimum significant differences were calculated with the Holm–Sidak test ($\alpha = 0.05$) using the Sigma Stat package (Systat Software inc., San Jose, CA, USA). The resistance factor (Rf) was calculated by dividing the determined GR₅₀, LD₅₀ or I₅₀ value of the resistant populations by that of the susceptible population to quantify the level of resistance.

3 RESULTS AND DISCUSSION

3.1 In vivo dose–response assays

In vivo dose–response assays showed that both resistant populations (R1 and R2) were highly cross-resistant to all ALS inhibitors

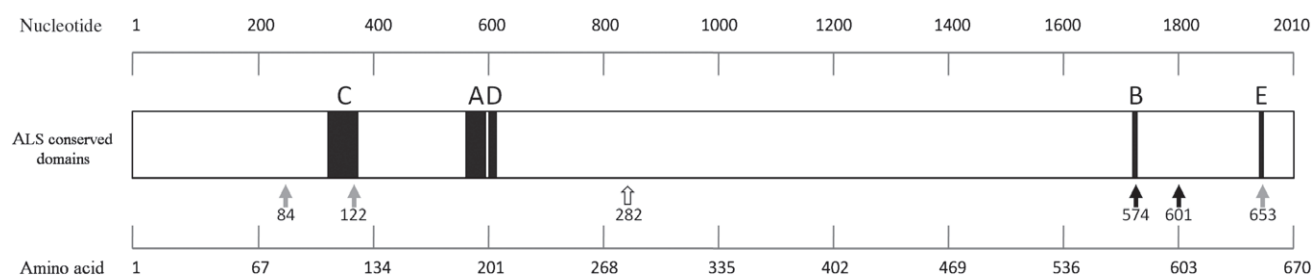


Figure 3. Scheme of the amino acid substitutions found in the *als* gene from R populations of *A. palmeri*. Black, gray and empty arrows indicate substitutions found in R1, R2 and in both of them, respectively. Only previously reported substitutions (at positions 122, 574 and 653) fall in the conserved domains of the ALS protein (named A to E). Amino acids are numbered according to the *A. thaliana* ALS sequence.

tested (imazethapyr, chlorimuron-ethyl and diclosulam), as shown in Figs 1 and 2. The resistance factors (Rfs) were >9.3 and >11 for GR_{50} and LD_{50} , respectively (Table 2).

Plant survival percentage at the recommended label rate (1X) was around 100% of that of the untreated control for all the herbicides (Fig. 2). Simultaneously, the remaining biomass of R plants at dose 1X ranged from 80 to 100% of that of the untreated control, except for the imazethapyr curve in R2, which showed a remaining biomass value of $\sim 45\%$ at the recommended rate (Fig. 1).

Although there is not a substantial difference between the R1 and R2 plant survival curves, the biomass reduction effects were stronger for R2 than for R1 for the three herbicides, as shown in Figs 1 and 2. Thus, the two variables provided complementary information to compare the resistance levels of R1 and R2.

Interestingly, in the absence of herbicides, both populations displayed significantly higher biomass values compared with the susceptible population (200% and 155% for R1 and R2, respectively).

The results presented herein confirm that the R1 and R2 populations of *A. palmeri* exhibit cross-resistance to three active ingredients belonging to different chemical families: IMIs, SUs and TPs.

3.2 In vitro enzyme activity assays

There were no significant differences in the levels of *in vitro* ALS activity between susceptible and resistant populations in the absence of herbicides. The specific activity of untreated controls ranged between 0.02 and 0.04 U mg^{-1} , similarly to values obtained for other plant crude extracts.^{32,33}

Experiments were set up to include three replicates at different concentrations (up to seven data points) in order to have a good distribution within the curve fitting. Based on I_{50} values (Table 3), both R populations had ALS isoforms with increased resistance to the three herbicides tested, compared with the S population. According to Rf values, these effects were very large for imazethapyr (Rf ranging from 30- to 40-fold), moderate for chlorimuron-ethyl (Rf ranging from 5- to 10-fold) and small, but still significant, for diclosulam (around 2.5-fold).

These results strongly support the hypothesis of a target-site resistance (TSR) mechanism in these *A. palmeri* resistant populations.

3.3 Molecular insight into ALS resistance

The study of the sequences of individual clones allowed a comparative analysis of the variability between R and S populations and within each population. All S population ALS sequences had 100% amino acid identity between them, whereas for the R1 and R2 populations, several clones presented mutations causing amino acid

substitutions. All substitutions found in the complete open reading frame are listed in Table 4 as well as the accession numbers of the complete sequences added to the NCBI database.

We identified four and five allelic versions in the R1 and R2 populations, respectively (Table 4). Even though we worked with genomic bulks, this represents a high number, considering that *Amaranthus* spp. are diploid plants that contain one copy of the *als* gene.³⁴ However, strong selection pressure for resistance to ALS inhibitors, a variety of amino acid changes that can confer resistance, and the outcrossing nature of *A. palmeri* could all contribute to the high number of polymorphisms we observed.

In the R1 population, the replacement of a tryptophan by a leucine at position 574 of the polypeptide chain was found in two clones. As a consequence of the later adoption of TP herbicides, this substitution has been more frequently associated with resistance to SUs and IMIs in a great variety of weed species,³⁶ but there are some reports that also associate such a substitution with resistance to TP.^{37,38} Thus, the W574 L substitution itself could explain the cross-resistance observed in the R1 population.

Tardif *et al.*³⁹ found that the W574 L mutation appears to have several pleiotropic effects on the growth and development of *Amaranthus powelli* plants. However, the absence of a biomass reduction and/or differential ALS activity under non-selection pressure between R and S populations does not provide any clue to fitness cost in our study. Nevertheless, a much more complex methodological framework is necessary to unequivocally associate a fitness cost to the resistant trait, as has been exhaustively reviewed by Vila-Aiub *et al.*⁴⁰

In the R2 population, the previously reported substitution S653 N was found in five clones. This mutation has been almost exclusively linked to IMI resistance, conferring low resistance to SUs in some cases.^{36,41} Interestingly, a new substitution that replaces an alanine by a serine at position 122 appeared in two clones. Although several replacements at this position (122) have been reported to confer resistance in plants, a serine has been found only in yeasts until now.⁴² As none of the known substitutions present in the R2 population justifies the high level of resistance to diclosulam, a contribution of the new substitutions found and/or a non-target-site resistance (NTSR) mechanism, mainly to TP resistance, cannot be discounted in this population.

Although W574 L triggers broad cross-resistance to all ALS-inhibiting herbicide families with higher magnitudes than S653 N,⁴³ the higher Rf from *in vivo* assays in the R1 over R2 populations cannot be wholly explained by W574 L itself, as the R2 population seems to have a higher frequency of plants with a target-site mutation (Table 4). Moreover, *in vitro* assays showed no significant differences between the two resistant populations,

suggesting a likely contribution of an NTSR mechanism in the R1 population.

Three unreported substitutions, M601I (R1), P84H (R2) and A282D (R1 and R2), were found outside the five ALS conserved domains, which were previously associated with TSR⁴⁴ (Fig. 3). Although it is more likely that they represent natural genetic variation, it remains to be elucidated if they contribute to the resistant phenotypes.

These molecular results show, for the first time, the presence of a TSR mechanism in *A. palmeri* Argentinian populations for resistance to ALS inhibitors.

4 CONCLUSIONS

We have confirmed the suspected cross-resistance to ALS-inhibiting herbicides in two populations of *A. palmeri* found in Argentina. This is the first report of TSR to ALS-inhibiting herbicides for this species in this country. We have found two amino acid substitutions in ALS from the resistant populations that were previously reported as responsible for TSR in plants: W574L (R1) and S653N (R2). Interestingly, the substitution A122S, only reported in yeasts up to now, has been found for the first time in plants (R2 population). The *in vivo* characterization of each purified homozygote progeny together with the heterologous expression of each *als* allele will be useful to evaluate the individual contribution of each substitution to the whole resistance phenotype.

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